

THAW A CREATURE

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Implementation Constants & Content Configuration - Canonical v0.9

Purpose: Exact first-pass values, content tables, implementation constraints, and AI guardrails

Design Source: Concise GDD v0.5

Prototype Source: Bare-Minimum Prototype Technical Specification v0.9

Launch Source: Technical Implementation Specification v0.5

Development Source: Development Plan v0.9

Status: Numerical values are first-pass balancing values unless explicitly structural

Revision notes

- Adds the canonical inherited Ice/creature size-roll system with `MaxVisualScale = 5.0`.
- Supersedes fixed rarity-only Pen Income/Speed with species baselines multiplied by copy Scale.
- Preserves 1-second aggregate Pen income and records Epic/Legendary economy tier values as real configuration.
- Rebalances prototype Warmth into big-number progression with exponential Recommended Warmth and regional cold multipliers.
- Expands prototype Hearth progression to a first-pass six-level big-number curve.
- Records percentage-based Warmth warning thresholds and scalable Hearth refill.
- Replaced the Week-1 prototype fixed `+30 sec` Thaw Surge with `GreatFrostThawMultiplier = 10`: the multiplier is active only during Great Frost, applies to Ice deposited mid-Frost for its remaining overlap, returns to `1x` during Day, and preserves Frost-earned progress.
- Synchronized sources to Prototype Technical Specification v0.9 and Development Plan v0.9.
- Added exact first-pass constants and AI guardrails for the Day-5 Survival Pressure Experiment.
- Kept the experiment prototype-only: no launch combat/enemy/pet-Vitality scope is implied unless the Week-1 promotion gate approves a specific subset and later canonical launch documents are revised.

- Integrated the approved Day-4 Stop-Gate Week-1 balance overrides: prototype carry multiplier `2.00x` and prototype creature speeds `16 / 22 / 28 / 34`.
 - Resolved previously flagged player-facing ambiguities instead of leaving them for AI inference.
 - Added exact shared wall-clock Great Frost phase/boot rules and Grab locking.
 - Added Special Ice origin/expiry behavior and launch onboarding-Ice configuration.
 - Added canonical reusable SilhouetteProfile mapping.
 - Normalized Creature/Rarity/Ice/WorldCycle configuration schema names across all documents.
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1. Document Authority

When documents appear to conflict, use this order: 1. Concise GDD - intended player experience and design. 2. Relevant Technical Specification - system behavior and architecture. Prototype Technical Specification governs Week-1 prototype scope. Launch Technical Specification governs production launch implementation. 3. This Constants Document - exact first-pass values and content configuration. 4. Development Plan - implementation order and production priority.

A later document version supersedes an earlier version of the same document. If an exact number changes during playtesting:

Update the configuration value. Do not rewrite the gameplay architecture merely to rebalance a number.

2. General Implementation Rule

All tunable gameplay values should live in configuration modules rather than being scattered through service/controller code. When implementing from this document:

Do not invent missing mechanics. If a number is not specified: 1. preserve the documented system; 2. choose the smallest safe technical implementation; 3. flag meaningful gameplay ambiguity; 4. do not create a new progression system to solve it.

3. AI IMPLEMENTATION CONSIDERATIONS

This section exists specifically to prevent ChatGPT, Claude, or another implementation assistant from making reasonable-looking assumptions that conflict with intentional design decisions. These rules apply to all implementation tasks.

3.1 Do Not Infer Missing Gameplay Design An AI must distinguish between four cases.
Specified Behavior Example:

Carrying Ice increases Warmth drain. Implement exactly as documented.

Configurable Value Example:

CarryDrainMultiplier = 1.75. Use the current canonical value unless the task is specifically a balance change.

Unspecified Technical Detail Example:

exact internal helper function names.

Choose the smallest clean implementation that preserves documented behavior. No clarification is normally required.

Unspecified Gameplay Behavior Example:

whether players should be allowed to intentionally drop carried Ice. Do not invent the answer. Flag the ambiguity before introducing that behavior.

3.2 Do Not Use Genre Conventions to Fill Design Gaps Do not assume that because another Roblox simulator commonly uses a mechanic, Thaw A Creature should also use it. Examples of prohibited assumptions:

“Warmth probably needs a maximum cap.”

“Pets probably need multiple stats.”

“There should probably be Thaw Slots.”

“Night probably makes the wilderness more dangerous.”

“Carrying a large Ice should probably slow the player.”

“Special Ice should probably guarantee Legendary.”

“Recommended Warmth probably means required Warmth.” All of those conflict with intentional decisions in the current design.

3.3 Never “Improve” Gameplay Design Without Approval An AI implementation assistant must not independently: add mechanics;

remove mechanics; combine progression systems; split progression systems; normalize intentionally exaggerated values; add monetization; add convenience features; add conventional simulator systems; because they seem: cleaner; more scalable; more balanced; more familiar; more monetizable; more realistic. Implementation and game-design revision are separate tasks.

3.4 Preserve System Responsibilities The following division is intentional. MaxWarmth

Expedition endurance. Determines how long the player can survive away from safety.

Hearth

MaxWarmth training speed. A stronger Hearth does not define the player's maximum Warmth.

Equipped Creature

Movement Speed. A stronger creature lets the player use their Warmth more efficiently by crossing distance faster.

Pen Creature

Passive Cash generation. A creature assigned to the Pen earns income. An Equipped creature does not simultaneously earn Pen income.

Ice

Mystery + retrieval opportunity. The player sees something desirable before knowing the exact result.

Thawing

Anticipation + real-time progression. It does not become a tapping or skill system.

Great Frost

World refresh + online thaw acceleration + recurring spectacle + Special Ice cadence. It is not primarily a punishment system. Do not merge these responsibilities unless an explicit design revision says to.

3.5 No Hidden Progression Gates RecommendedWarmth means:

player guidance. It does not mean:

entry requirement. Never implement:

```
1 if player.MaxWarmth < region.RecommendedWarmth then
```

```
2     denyEntry()  
3 end
```

Players may attempt any region. Practical viability creates progression pressure. Permission does not.

3.6 Unlimited Means Unlimited at the Gameplay Level The following are intentionally uncapped:

MaxWarmth and:

logical ThawingIce count Do not derive gameplay caps from: Hearth Level; UI capacity; display anchors; Pen geometry; implementation convenience; model limits. Technical safety limits may exist only as exploit/performance protection. They must not function as normal progression.

3.7 Visual Structures Are Not Automatically Gameplay Structures Examples:

12 Ice display positions != maximum 12 thawing Ice.

Region Parts != locked zones.

Four Pen creature anchors != four total owned creatures.

Candidate Ice spawn points != fixed spawn order.

Five Hearth visual models != five mechanical Hearth levels. Do not infer gameplay rules from Workspace hierarchy or presentation assets.

3.8 Hidden Reward Information Must Remain Hidden Before final reveal, the exact reward must not be deliberately exposed to the client unless required by an approved presentation system. Server-only reward data includes: CreatureID; exact rarity; reward value. Client-safe information may include: Ice Type; approved silhouette profile; thaw percentage; visual thaw stage; approved rarity foreshadow effect. Do not accidentally expose hidden outcomes through: replicated object names; obvious Attributes; UI text; model names; client-readable config records; different timers based on rarity.

3.9 Thaw Completion Is Not Creature Completion When:

```
1 ThawProgress >= 100%
```

the Ice becomes:

READY TO CRACK It does not automatically grant the creature. Creature acquisition occurs only after:

CRACK -> CRACK -> SMASH and successful server resolution of the final interaction.

3.10 Offline Progress Rule The Week-1 prototype has no persistence, so offline progress is not evaluated there.

For launch, offline players receive normal real-time thaw progress. They do not receive historical one-time launch Great Frost Thaw Surges that occurred while they were absent. Do not calculate:

number of missed Great Frosts x launch Thaw Surge

on rejoin.

The intended launch distinction is:

offline = still useful

online = better

3.11 Great Frost Reset Rule At Great Frost: Reset available/unclaimed wilderness Ice. Preserve carried Ice; Pen Ice; unfinished thawing Ice; Ready Ice;

creatures; player progression.

Apply the currently configured Great Frost thaw acceleration without inventing extra penalties.

Prototype override: - Day thaw rate = 1x ; - Great Frost thaw rate = 10x ; - the multiplier applies only for each ThawItem's real-time overlap with the active Frost interval, including Ice deposited during Frost; - Day resumes at 1x ; - Frost-earned progress is preserved.

Launch behavior remains the one-time percentage-based Thaw Surge defined in Section 32 until a later launch design revision explicitly promotes the prototype multiplier.

3.12 Great Frost Is Opportunity, Not Punishment Great Frost should not automatically: kill the player; reduce Warmth; teleport the player; destroy carried Ice; require shelter; turn the world into a generic enemy wave; apply a survival debuff. Its current launch role is:

refresh opportunities + accelerate thawing + occasionally create Special Ice + deliver spectacle.

Prototype exception: If the approved Day-5 Survival Pressure Experiment is active, Great Frost may clear/rebuild the authored Guarded Ice encounters together with the wilderness Ice field. This does not make Great Frost itself a punishment or enemy phase.

3.13 Speed Progression Is Intentionally Large Do not normalize:

16 -> 20 -> 24 -> 28 -> 32 into conservative increments merely because they look large. The player should visibly feel:

"I am much faster now." This is a core progression fantasy. Balance testing may later change the numbers. Implementation should not preemptively weaken them.

3.14 Carrying Ice Does Not Slow Movement The cost of carrying Ice is:

increased Warmth drain. Do not also automatically apply: WalkSpeed reduction; jump reduction; stamina penalty; heavier controls. Doing so would overlap the intended risk system.

3.15 Creature Stats Must Remain Simple Creature copies may differ because of their inherited size lineage.

Approved per-copy variation is limited to one inherited size roll from the originating Ice:

- WeightMultiplier ;
- derived Scale ;
- derived FinalIncome ;
- derived FinalWalkSpeed .

These are all consequences of the same inherited WeightMultiplier . There is no independent random roll for Income or Speed.

Do not add unless a future explicit design revision approves them:

- random independent Income rolls;

- random independent Speed rolls;
- Warmth bonus;
- Luck;
- Damage;
- Defense;
- levels;
- XP;
- affixes;
- traits;
- skills;
- stamina;
- carrying capacity;
- random stat points;
- traversal abilities;
- additional pet-stat systems;
- rarity multipliers beyond configured systems.

Design principle:

Creature copies may differ because of their inherited size lineage. Do not turn creatures into generic randomized-stat RPG pets.

Prototype exception: During the exact approved Day-5 Survival Pressure Experiment only, the currently Equipped creature may have temporary session-only `CreatureVitality` plus a boolean Frosted/Downed state. This is threat-state for the experiment, not a permanent creature stat or launch identity. It must never alter ownership, create Damage/Defense/levels, affect Pen creatures, or become a generic creature-stat framework.

3.16 Special Ice Does Not Automatically Mean Legendary Special Ice means:

high-value opportunity with elevated rarity.

Use the configured Special Ice rarity tables. Do not replace them with:

1 SpecialIce = GuaranteedLegendary

unless a future design change explicitly says so.

3.17 Do Not Add Convenience Systems Without Approval Do not independently introduce: teleport home; sprint; mounts; dash; auto-sell; auto-thaw; auto-open; extra Ice storage; raw-Ice backpack; fast travel; pet storage upgrades; Luck; quests; rebirth; Daily rewards. A convenience mechanic may accidentally bypass the game's intended friction.

3.18 Protect the USP Before implementing any inferred behavior, ask whether it weakens one of these three experiences:

I saw something I desperately wanted.

I barely managed to bring it home.

I couldn't wait to discover what was inside. If it does:

do not implement it without explicit approval.

3.19 Protect the Five Hero Moments Implementation decisions must not accidentally undermine: 1. First Great Frost. 2. Impossible-Looking Ice. 3. Barely-Made-It-Home. 4. Legendary Smash. 5. Becoming the Endgame Player. Examples: Teleporting home while carrying Damages Hero #3. Automatically granting a completed thaw reward Damages Hero #4. Weak Speed differences Damages Hero #5. Hiding future regions completely Damages Hero #2. Treating Great Frost as a minor UI timer Damages Hero #1.

3.20 Configuration Is Not Permission to Rebalance A number being stored in config means:

it should be easy for the designer to change. It does not mean the implementation assistant may change it during an unrelated task.

Unless the current task is specifically balancing:

use the canonical value.

3.21 Do Not Resolve Document Contradictions Silently If canonical documents appear to conflict: 1. identify the conflicting instructions; 2. apply the Document Authority order where the resolution is clear; 3. if meaningful ambiguity remains, stop and flag it; 4. do not silently invent a hybrid rule.

3.22 Prototype Overrides Are Not Launch Design Changes Prototype constants intentionally compress: thaw times; Great Frost timing; Hearth progression; content. These are testing overrides. Do not copy them into launch merely because they already exist in code. The architecture should support:

1 PrototypeConfig

versus:

1 LaunchConfig

or equivalent clean configuration separation.

3.23 Prototype Omission Does Not Mean Launch Removal If a launch mechanic is missing from the Week-1 prototype, that means:

not required for prototype validation not:

removed from the game. Examples: persistence; offline thawing; multiplayer; Special Ice; Legendary creatures; Ancient/Black/Meteor Ice. Consult the Launch Technical Specification before interpreting prototype omissions.

3.24 Smallest Robust Implementation When a purely technical choice is unspecified:

implement the smallest robust solution that preserves the current architecture. Avoid speculative infrastructure such as: generalized ability framework; quest framework; universal stat framework; complex item framework; generic combat framework; external framework dependency; unless current requirements actually need it.

3.25 AI Ambiguity Protocol If an implementation task encounters an unspecified decision, classify it. Purely Technical and Behavior-Preserving Proceed. Example:

helper module naming.

Tunable Number Already Represented by Existing Config Use the current config/default.

Player-Facing Gameplay Consequence Do not assume. Flag it. Examples: introducing a new failure penalty; adding trading; adding paid progression modifiers; auto-routing behavior not specified anywhere; any new player-facing shortcut that bypasses retrieval risk.

3.26 Resolved Player-Facing Edge Decisions The following questions are no longer ambiguous.

Intentional raw-Ice drop Not available at prototype or launch. A carry ends through Pen deposit or documented failure/reset/death/disconnect handling.

Freeze / normal respawn recovery Return to own base and set CurrentWarmth = MaxWarmth immediately. Permanent progression is preserved.

Launch base topology Up to four personal plots are grouped side-by-side along one shared home-camp edge. All plots face the same primary wilderness direction.

Visitor Hearth behavior Another player's WarmZone does not refill CurrentWarmth and does not train MaxWarmth.

First-time onboarding Ice Enabled at launch only. A player with no discovered creatures, no ThawingIce, and no carried Ice may receive one private Regular Ice at their authored OnboardingIceSpawnPoint. It uses normal Regular random reward rules, is not contestable, and is not reset by Great Frost.

Special Ice lifecycle Unclaimed Special Ice expires at the next Great Frost field reset. If a carrier fails before deposit, return it to its origin only while its OriginCycleIndex opportunity is still current; otherwise remove it. Carried Special Ice still survives Great Frost like any carried Ice.

Great Frost server boot Boot during Day: populate current field immediately. Boot during Great Frost: wait until next Day. Do not retroactively award the launch Thaw Surge or Special Ice from a Frost transition the server did not observe.

3.27 Approved Prototype Exception - Day-5 Survival Pressure The Development Plan v0.9 and Prototype Technical Specification v0.9 explicitly authorize the EXP-D5 Survival Pressure Experiment. Within those tasks only, AI/developers may implement: - one universal swing tool; - one stationary Frost Sprite enemy type; - selected authored Guarded Ice encounters; - player enemy damage as direct CurrentWarmth loss; - temporary session-only Equipped-creature Vitality/Frosted state; - minimal survival UI/feedback; - breakable environmental obstacles only after the first survival validation gate passes/partially passes.

This exception does not approve: - Player HP; - permanent creature death/loss; - creature combat AI; - permanent combat stats; - multiple weapons; - weapon rarity/durability/upgrades/inventory; - enemy drops/currency/XP/levels; - roaming/pathfinding enemies; - bosses; - material currencies; - crafting/recipes; - generalized combat/item/ability/resource frameworks.

If an EXP implementation requires a player-facing behavior not specified by the Plan, Prototype Spec, or this Constants document, do not infer it from survival-game conventions. Flag it.

3.28 Experiment Does Not Equal Launch Promotion The launch documents remain authoritative for launch scope. The presence of experiment code or config is not evidence that combat/enemies/pet Vitality belong in production. At the End-of-Week-1 gate, record PASS, PARTIAL PASS, or FAIL. Only a specifically promoted subset may enter later GDD/Launch Technical/Development Plan revisions. If FAIL, remove/disable experiment-only gameplay before Week 2.

4. Deprecated Rules - DO NOT IMPLEMENT

The following belong to older versions of the design and are no longer canonical: NO: Warmth Lv.1-10 as the player's progression stat. NO: Fixed Warmth capacity granted directly by upgrades. NO: Equipped creature adding Warmth. NO: Dedicated Thaw Machine. NO: Limited Thaw Slots. NO: Individual Ice respawning every 6/8/10/12/15 seconds. NO: Creature-specific traversal abilities. NO: Separate Speed upgrade tree. NO: Speed treadmill. NO: Carry movement slowdown. Current model:

MaxWarmth = uncapped trainable stat
Hearth = Warmth training speed
Equipped creature = Movement Speed
Pen = creatures + thawing Ice
Great Frost = primary world Ice refresh

5. Structural Rules

These rules are not casual balancing variables. Player 1 Equipped creature maximum. 1 carried Ice maximum. 4 active Pen creature positions. Unlimited owned creature counts. Unlimited logical thawing Ice. 1 currency: Cash. MaxWarmth has no cap. Ice Wilderness Ice reward is rolled server-side. One wilderness Ice may have only one owner. Ice cannot enter normal Inventory before thawing. Thaw duration depends on Ice Type, not hidden rarity. Areas No hard Warmth gates. Regions provide Recommended Warmth only. Great Frost resets only available/unclaimed wilderness Ice; carried Ice survives; Pen Ice survives; Ready Ice survives; prototype Pen Ice receives the active 10x Great Frost Thaw Boost; launch online players receive the configured launch Thaw Surge.

6. Player / Server Constants

Launch

Constant	Value
Target Players Per Server	4
Base WalkSpeed with no creature	16
Active Pen Creature Slots	4
Equipped Creature Limit	1
Carried Ice Limit	1
Currency	Cash

Prototype

Constant	Value
MaxPlayers	1
Active Pen Creature Slots	4
Equipped Creature Limit	1
Carried Ice Limit	1

7. Warmth - Core / Launch Direction

The big-number Warmth direction is now canonical, not a prototype-only visual trick.

Constant	Initial Value
Starting MaxWarmth	100
Spawn CurrentWarmth	MaxWarmth
Base Wilderness Drain	4 Warmth/sec
Carry Drain Multiplier	2.00x
Hearth Refill	50% of current MaxWarmth / sec
Low-Warmth Warning	35%
Critical-Warmth Warning	15%
Frozen Threshold	0

Region cold severity multiplies the base drain and provides the exponential progression curve. Exact region multipliers and Recommended Warmth values are defined in Section 11.

8. Warmth Training

While inside the player's own Hearth:

```
MaxWarmth += HearthTrainingRate * dt
```

Warmth training: - automatic; - uncapped; - no XP; - no training currency; - no player input required.

The revised progression intentionally favors satisfying whole-number growth and very large long-term values.

CurrentWarmth refill scales with MaxWarmth:

```
CurrentWarmthRefillPerSecond = MaxWarmth * 0.50
```

This preserves the old prototype feel of roughly filling from empty in about two seconds even after MaxWarmth becomes enormous.

9. Hearth Progression - Revised Prototype First Pass

The initial 3-level Day-5 Hearth proved the mechanic. Days 6-7 expand it.

Hearth Lv.	Upgrade Cost	Warmth Training Rate
1	Start	+10/sec
2	\$150	+50/sec
3	\$1,000	+250/sec
4	\$7,500	+1,250/sec
5	\$60,000	+6,250/sec
6	\$500,000	+31,250/sec

These are first-pass prototype progression values and should be tuned from actual pacing. The functional rule is structural: - Hearth upgrade never grants a fixed MaxWarmth amount; - Hearth level only changes ongoing training rate; - MaxWarmth remains uncapped.

The former Day-5 `+0.50 / +1.00 / +1.75` rates are superseded once D7-R03 is implemented.

10. Hearth Visual Stages

Prototype presentation target:

Lv.1	Small Hearth
Lv.2	Strong Hearth
Lv.3	Iron Brazier
Lv.4	Runic Hearth
Lv.5	Frostfire Forge
Lv.6	Great Runic Furnace

These do not require six unique production meshes. Presentation may reuse geometry with stronger flame/light/runes/scale.

Always communicate:

```
HEARTH LV. X  
TRAINING +<rate> WARMTH/s  
NEXT $<cost> / MAX
```

11. Region Recommended Warmth + Cold Severity

First-pass big-number progression:

Region	Main Ice	Recommended Warmth	Cold Multiplier
Snowfield	Regular	100	1x
Frozen Pass	Thick	1,000	10x
Ancient Expanse	Ancient	10,000	100x
Black Ice Hollow	Black	100,000	1,000x
Meteor Reach	Meteor	1,000,000	10,000x

Guidance only. Never a server access condition.

Warmth drain:

```
WarmthDrainPerSecond =  
    BaseWarmthDrain  
    * RegionColdMultiplier  
    * (Carrying and CarryMultiplier or 1)
```

This makes the numbers grow by orders of magnitude while keeping the physical world manageable.

12. World Distance Targets

Keep current first-pass route anchors unless level-design playtesting changes them:

Region	Target Route Distance
Snowfield	60-100 studs
Frozen Pass	180-240 studs
Ancient Expanse	340-460 studs
Black Ice Hollow	580-760 studs
Meteor Reach	900-1,100 studs

Recommended Warmth is not calculated from distance alone; creature Speed and regional cold combine to create risk.

13. Warmth Core Constants - Revised Prototype

Starting MaxWarmth	100
Base Warmth Drain	4/sec
Carry Multiplier	2.00x
Hearth Refill	50% of MaxWarmth / sec
Low Warmth Threshold	35% of MaxWarmth
Critical Warmth Threshold	15% of MaxWarmth

The regional cold multiplier supplies the dramatic progression scaling. CurrentWarmth is still restored to full on canonical Freeze recovery.

14. Creature Size / Weight Jackpot System

Each spawned Ice gets exactly one server-authoritative `WeightMultiplier`.

For any Ice/creature with `ReferenceWeight`:

```
ActualWeight = ReferenceWeight * WeightMultiplier
Scale = cbrt(ActualWeight / ReferenceWeight)
Scale = cbrt(WeightMultiplier)
```

Supported visual Scale:

```
Minimum Visual Scale    0.75x
Maximum Visual Scale    5.00x
```

The runtime must roll `WeightMultiplier`, then derive `Scale` through the formula. Do not separately reroll visual `Scale`.

Size bands

Band	Chance	Visual Scale	WeightMultiplier Range
Small	8.0%	0.75-0.95x	0.422-0.857x
Normal	72.0%	0.95-1.25x	0.857-1.953x
Large	15.0%	1.25-1.75x	1.953-5.359x
Huge	4.0%	1.75-2.50x	5.359-15.625x
Colossal	0.9%	2.50-3.50x	15.625-42.875x
Absurd	0.1%	3.50-5.00x	42.875-125.000x

Within each band, bias the `WeightMultiplier` toward the band's lower bound so values near 5.0x visual `Scale` are dramatically rarer than simply entering the Absurd band. First-pass implementation:

```
t = Random() ^ 2
WeightMultiplier = lerp(BandMinWeight, BandMaxWeight, t)
```

A true near-5x copy is intended to be an exceptional social/jackpot event.

The `WeightMultiplier` is inherited unchanged:

```
wilderness Ice
= carried Ice
= thaw record
= revealed CreatureCopy
```

15. Ice Base Visual / Weight Contract

Each Ice type must define explicit:

```
ReferenceWeight
BaseVisualScale
```

The exact authored dimensions belong to the repository asset/config audit. Do not invent world-space dimensions if the existing visual asset already establishes its 1.0x baseline.

Recommended relative base visual multipliers for prototype tuning:

```
Regular    1.00
Thick      1.35
Ancient    1.75
Black      2.20
Meteor     3.00
```

Runtime visual scale:

```
FinalIceVisualScale = BaseVisualScale * SizeScale
```

All extreme Ice visuals should avoid blocking traversal physics.

16. Creature Base Stat Contract

Every configured species must define:

```
ReferenceWeight
BaseVisualScale
BaseIncomePerSecond
BaseWalkSpeed
```

Runtime copy values:

```
Scale = cbrt(WeightMultiplier)
IncomeMultiplier = Scale ^ 2
SpeedMultiplier = 1 + 0.15 * (Scale - 1)
FinalIncome = max(1, round(BaseIncomePerSecond * IncomeMultiplier))
FinalWalkSpeed = BaseWalkSpeed * SpeedMultiplier
```

At the supported extremes:

```
0.75x Scale -> SpeedMultiplier ~= 0.9625x
1.00x Scale -> 1.00x
2.00x Scale -> 1.15x
3.00x Scale -> 1.30x
4.00x Scale -> 1.45x
5.00x Scale -> 1.60x
```

Income intentionally scales much harder:

```
1x Scale -> 1x income
2x Scale -> 4x income
3x Scale -> 9x income
4x Scale -> 16x income
5x Scale -> 25x income
```

Do not assign the 11 current species' exact base values by AI guess. D6-R00 must audit the real repository model IDs; base stat/content mapping is then explicitly approved.

17. Supported Rarity / Economy Tiers

Configured economy tier labels:

1. Common
2. Uncommon
3. Rare
4. Epic
5. Legendary

Current Pen-income reference baselines retained for migration/comparison only:

Common	\$2/sec
Uncommon	\$5/sec
Rare	\$12/sec
Epic	\$20/sec
Legendary	\$28/sec

After D6-R05/D6-R06, actual Pen income comes from species `BaseIncomePerSecond * Scale^2`, not directly from rarity.

The existence of Epic/Legendary economy tiers does not automatically define prototype creature content, pools, or weights.

18. Creature Stat Presentation

Every visible creature copy should expose at minimum:

```
SPEED <rounded FinalWalkSpeed>
PEN $<FinalIncome>/s
```

An Equipped creature displays its intrinsic `PEN $/s` value for comparison but earns zero while Equipped.

Extreme copy Scale should remain the primary visual signal; text is secondary.

19. Pen Income

Income interval:

```
1 second
```

Server calculates one aggregate payment:

```
TotalIncome =  
  PenSlot1FinalIncome  
  + PenSlot2FinalIncome  
  + PenSlot3FinalIncome  
  + PenSlot4FinalIncome
```

No independent timer per creature. No offline Pen income. Equipped and Inventory copies earn zero.

20. Ownership / GUID Structural Constants

Max Equipped Copies	1
Active Pen Creature Slots	4
Creature identity key	CreatureGUID
Discovery identity key	CreatureID / species

Persistent copy payload should remain minimal:

```
CreatureID  
WeightMultiplier
```

Scale / Income / Speed are derived.

21. Launch Creature Roster

Common 1. Penguin 2. Rabbit 3. Seal 4. Arctic Fox

Uncommon 5. Wolf 6. Polar Bear 7. Moose 8. Walrus

Rare 9. Mammoth 10. Sabertooth 11. Woolly Rhino 12. Raptor 13. Dire Wolf

Legendary 14. Yeti 15. Frost Serpent 16. Ice Dragon

Total:

16

22. Prototype Creature Roster

Common Penguin Rabbit

Seal **Uncommon** Wolf Polar Bear **Rare** Mammoth Sabertooth Raptor Total:

8

23. Normal Ice Rarity Weights

Regular Ice

Rarity	Weight
Common	75%
Uncommon	23%
Rare	2%
Legendary	0%

Thick Ice

Rarity	Weight
Common	45%
Uncommon	45%
Rare	10%
Legendary	0%

Ancient Ice

Rarity	Weight
Common	10%
Uncommon	40%
Rare	45%
Legendary	5%

Black Ice

Rarity	Weight
Common	0%
Uncommon	20%
Rare	55%
Legendary	25%

Meteor Ice

Rarity	Weight
Common	0%
Uncommon	0%

Rare	60%
Legendary	40%

24. Launch Creature Pools

Once rarity has been rolled:

choose uniformly among eligible creatures of that rarity within the current Ice pool. Regular **Common**

Penguin Rabbit Seal Arctic Fox **Uncommon** Wolf Polar Bear **Rare** Mammoth **Legendary** None

Thick **Common** Seal Arctic Fox **Uncommon** Wolf Polar Bear Moose Walrus **Rare** Mammoth Sabertooth **Legendary** None

Ancient **Common** Arctic Fox

Uncommon Polar Bear Moose Walrus **Rare** Mammoth Sabertooth Woolly Rhino Raptor **Legendary** Yeti

Black **Common** None **Uncommon** Polar Bear Moose **Rare** Sabertooth Raptor Dire Wolf **Legendary** Yeti Frost Serpent

Meteor **Common** None **Uncommon** None

Rare Woolly Rhino Raptor Dire Wolf **Legendary** Yeti Frost Serpent Ice Dragon

25. Prototype Ice Pools

Regular **Common** Penguin Rabbit Seal **Uncommon** Wolf Polar Bear **Rare** Mammoth Weights:

75 / 23 / 2

Thick **Common** Seal **Uncommon** Wolf Polar Bear **Rare**

Mammoth Sabertooth Raptor Weights:

45 / 40 / 15 Prototype Thick deliberately exposes Rare more often than launch Thick. This exists to accelerate validation.

26. Bad-Luck Protection

Initial launch setting:

DISABLED

```
1 BadLuckProtectionEnabled = false
2 CommonStreakThreshold = 4
```

The reward currently belongs to shared wilderness Ice before player ownership. Player-specific pity therefore does not cleanly fit the current spawn architecture. Do not implement until explicitly re-enabled.

27. Ice Spawn Counts - Launch

Initial four-player target:

Ice	Active Per Great Frost
Regular	12
Thick	8
Ancient	8
Black	6
Meteor	4
Total	38

Ice remains until: claimed; or removed by the next Great Frost. No independent normal respawn timer.

28. Candidate Spawn Point Targets

Ice	Candidate Points	Active
Regular	18	12
Thick	12	8
Ancient	12	8
Black	9	6
Meteor	6	4

The game selects among authored positions. It does not generate arbitrary world coordinates.

29. Prototype Spawn Counts

Ice	Candidate Points	Active
Regular	6	4
Thick	4	2

30. Great Frost - Launch Timing

Constant	Value
Full Cycle Length	300 sec / 5 min
Normal Day Portion	285 sec
Great Frost Duration	15 sec

Launch should use shared wall-clock cycle timing.

CycleEpochUnix = 0. Canonical phase rule: CyclePosition = (CurrentUnixTime - CycleEpochUnix) % CycleLength. Day when CyclePosition < 285. Great Frost when CyclePosition >= 285.

GrabLockedDuringGreatFrost = true. A server tracks LastProcessedFrostCycle at runtime so the same Frost transition cannot be processed twice. A server booting mid-Frost does not retroactively award Thaw Surge or Special Ice.

31. Great Frost Sequence

Within the 15-second Frost window: 0 sec Great Frost begins. ~2 sec Transition/world presentation begins. ~3 sec Remove available/unclaimed Ice. ~4 sec Apply online Thaw Surge. ~8 sec Populate new Ice field. ~10 sec Present Special Ice announcement if applicable. 15 sec Normal phase resumes. GrabIce confirmations become valid again when normal Day resumes. Presentation timing may change without altering gameplay rules.

32. Great Frost Thaw Surge - Launch

Each unfinished Ice belonging to a player currently online receives:
15% of that Ice’s BaseThawDuration Formula:

```
1 BonusProgressSeconds += BaseDurationSeconds x 0.15
```

Examples:

Ice	Base Timer	Frost Bonus
Regular	1 min	9 sec
Thick	3 min	27 sec
Ancient	7 min	63 sec
Black	15 min	135 sec
Meteor	25 min	225 sec

Ready Ice is unaffected. Offline players do not receive the bonus.

33. Prototype Great Frost Timing + Thaw Boost

Constant	Prototype
Cycle Length	120 sec
Normal Portion	105 sec
Great Frost Duration	15 sec
GreatFrostThawMultiplier	10x
Day Thaw Multiplier	1x

The prototype multiplier is deliberately exaggerated for validation.

Rules: - `GreatFrostThawMultiplier` is active only while `CurrentPhase == GreatFrost`; - unfinished Ice already thawing accelerates at `10x` for the Frost overlap; - Ice deposited during Great Frost immediately accelerates at `10x` from `DepositedAtTime` until Frost ends; - when Day resumes, thaw speed immediately returns to `1x`; - progress earned during Great Frost is preserved; - already-Ready Ice remains Ready; - this replaces the superseded prototype fixed `+30 sec` grant.

Conceptual active-Frost calculation:

```
1 OverlapStart = max(  
2     DepositedAtTime,  
3     GreatFrostStartedAtTime  
4 )  
5 FrostSeconds = max(0, CurrentTime - OverlapStart)  
6 FrostExtra =  
7     FrostSeconds  
8     * (GreatFrostThawMultiplier - 1)  
9 EffectiveElapsed =  
10    NormalElapsed  
11    + BonusProgressSeconds  
12    + FrostExtra
```

At Great Frost -> Day, preserve the earned extra progress by folding the completed Frost overlap into `BonusProgressSeconds` (or an equivalent authoritative accumulated-bonus field) before the live Frost contribution becomes zero.

33A. Prototype Survival Pressure Experiment - First-Pass Constants

These constants are active only while the `EXP-D5` experiment is enabled. They are intentionally simple first-pass values for rapid playtesting, not launch balance.

Universal survival tool:

Constant	Prototype Value
<code>ToolServerHitRange</code>	8 studs
<code>ToolSwingCooldown</code>	0.65 sec
<code>ToolDamageToFrostSprite</code>	1

Frost Sprite:

Constant	Prototype Value
<code>FrostSpriteMaxHealth</code>	3
<code>EngagementRadius</code>	26 studs
<code>AttackTelegraph</code>	0.75 sec
<code>AttackCooldown</code>	2.40 sec
<code>ProjectileSpeed</code>	40 studs/sec
<code>PlayerWarmthDamagePerHit</code>	10

CreatureVitalityDamagePerHit	25
TargetingRule	nearest valid target

Target locks when the telegraph begins.

Equipped creature experimental state:

Constant	Prototype Value
CreatureVitalityMax	100
FrostedAtVitality	0
FrostedSpeedBonusEnabled	false
HearthRecovery	full restore on Hearth entry

Hearth recovery means a valid entry into the player's Hearth/WarmZone immediately restores full experimental Vitality, clears Frosted, and restores the Equipped Speed bonus.

Guarded Ice authoring target for the first test:

Encounter Profile	Guardian Count
Light Guarded Regular	1 Frost Sprite
Medium Guarded Thick	2 Frost Sprites
Heavy Guarded Thick	3 Frost Sprites
Maximum Guarded Encounters	3 authored Ice opportunities

All remaining prototype Ice opportunities should remain unguarded. Do not add random enemy population or independent enemy respawn timers. Great Frost clears and rebuilds active guardians with the Ice field.

Conditional breakable-environment test - only after PASS/PARTIAL PASS:

Breakable	Hits To Break
Frozen Branch	3
Brittle Rock / Ice Rock	5

Breakables grant no Wood, Stone, currency, XP, drops, or crafting materials.

Balance note: These values are allowed to change during the survival playtest as explicit balance changes. Changing them does not authorize new mechanics.

34. Thaw Durations - Launch

Ice	Base Thaw Time
Regular	60 sec
Thick	180 sec / 3 min
Ancient	420 sec / 7 min
Black	900 sec / 15 min
Meteor	1,500 sec / 25 min

Timers continue offline.

Hidden rarity does not alter duration.

35. Prototype Thaw Durations

	Prototype Time
Ice	20 sec
Regular	20 sec
Thick	45 sec

These values are validation overrides only.

36. Thaw Progress States

Progress	Visual State
0-35%	Frozen / opaque
35-70%	Partial thaw
70-99%	Nearly Ready
100%	READY TO CRACK

Visual progress must not leak hidden rarity.

37. Offline Thawing

Launch formula:

```
1 Elapsed =  
2   CurrentUnixTime  
3   - DepositedAtUnix  
4   + BonusProgressSeconds
```

```
1 Progress =  
2   clamp(  
3     Elapsed / BaseDurationSeconds,  
4     0,  
5     1  
6   )
```

Offline:

normal elapsed time only.

Online:

normal elapsed time + any Great Frost bonuses actually received.

38. Special Ice Cadence

Initial launch value:

1 Special Ice every 3 Great Frost cycles At a five-minute cycle:
approximately every 15 minutes.

1 SpecialIceEveryNCycles = 3

39. Special Ice Spawn Rule

Special Ice: spawns in addition to normal population; modifies a normal Ice Type; uses a valid authored spawn position; is shared/contestable; follows normal carry rules. stores OriginCycleIndex and OriginSpawnPointID; expires from the shared field at the next Great Frost if still unclaimed; returns to its origin after carrier failure/disconnect only while its originating opportunity remains current; disappears on such a failure if a newer Great Frost has superseded that opportunity. Eligible Types:

Ancient / Black / Meteor Selection:

Type	Weight
Ancient	50%
Black	30%
Meteor	20%

40. Special Ice Rarity Weights

Special Ice guarantees:

Rare or Legendary Ancient Special

Rarity	Weight
Rare	75%
Legendary	25%

Black Special

Rarity	Weight
Rare	55%
Legendary	45%

Meteor Special

Rarity	Weight
Rare	35%
Legendary	65%

Creature remains restricted to that Ice Type's normal creature pool.

41. Special Ice Presentation

Minimum: visually unusual; visibly larger and/or runic; distinguishable at distance; server-wide notification. Recommended phrasing:

A Strange Ice Has Awakened

Do not reveal the creature.

42. Reveal Constants

Constant	Value
Required Hits	3
Hit Sequence	Crack / Crack / Smash
Minimum Accepted Hit Interval	0.20 sec
Target Total Interaction	3-6 sec
Failure	None
Accuracy Requirement	None
Combo Requirement	None

43. Rarity Foreshadowing

Common No special foreshadow required. **Uncommon** No special foreshadow required. **Rare** Hit #2 may use subtle: glow; sound. **Legendary** Hit #2 may use stronger: glow; particles; sound. Exact rarity is revealed on final Smash.

44. Silhouette Information Rule

Silhouette is a:

hint, not guaranteed exact identification. Desired progression:

vague at distance -> suggestive nearby -> clearer during thaw -> nearly recognizable near Ready -> exact identity on reveal Several creatures may share a silhouette profile.

Canonical launch SilhouetteProfile mapping:

Profile	Creatures
Compact	Penguin / Rabbit / Seal
AgileQuadruped	Arctic Fox / Wolf
HeavyBeast	Polar Bear / Walrus
Antlered	Moose

GiantAncient	Mammoth / Woolly Rhino
AncientPredator	Sabertooth / Raptor / Dire Wolf
MythicUnknown	Yeti / Frost Serpent / Ice Dragon

Profiles are deliberately coarser than exact species. Presentation may obscure/distort further at distance. Antlered currently maps only to Moose; that is acceptable because the global rule is hint-not-guaranteed-identification where practical, not mandatory ambiguity for every species.

45. Interaction Distances

Interaction	Max Server Distance
Grab Ice	14 studs
Deposit at Pen	20 studs
Begin Reveal	18 studs
Crack while revealing	20 studs
Hearth Upgrade	20 studs
Pen Management	25 studs
Equip / Sell	25 studs
EXP Survival Tool Hit	8 studs (prototype only)

These are server-validation distances. Prompt display distance may differ.

46. Request Rate Limits

Action	Minimum Interval
--------	------------------

GrabIce	0.35 sec
DepositCarriedIce	0.50 sec
BeginReveal	0.50 sec
Crack	0.20 sec
Equip / Unequip	0.25 sec
Pen Management	0.25 sec
Sell	0.25 sec
UpgradeHearth	0.50 sec
EXP Survival Tool Swing	0.65 sec (prototype only)

These are technical anti-spam protections, not intended gameplay cooldowns.

47. Pen Rules

Launch: 4 active earning creature positions; duplicates allowed; thawing Ice consumes zero creature positions; Pen creatures earn income; Equipped creature earns no Pen income; Sell from Pen leaves slot empty; Equip from Pen leaves slot empty; no automatic Inventory refill.

48. Inventory Rule

Inventory is derived from ownership. For each species:

```
1 InventoryCount =  
2   OwnedCount  
3   - PenCopies  
4   - EquippedCopies
```

Invariant:

```
1 PenCopies + EquippedCopies <= OwnedCount
```

Inventory is not a second ownership database.

49. New Reveal Routing

After reveal: Pen has space Place creature into first available active Pen position. Pen full Creature remains unallocated in Inventory. Display:

Pen Full - Added to Inventory No modal decision is required.

50. Equip Rules

Equip/Unequip occurs at the player's own base. Inventory -> Equip Creature becomes Equipped. Pen -> Equip Pen position becomes empty. Replace Equipped Old Equipped creature returns to Inventory. Unequip Creature returns to Inventory. No automatic Pen refill.

51. Sell Rules

Sell at own base. Inventory

```
1 OwnedCount -= 1  
2 Cash += SellValue
```

Pen

```
1 Clear Pen Slot  
2 OwnedCount -= 1  
3 Cash += SellValue
```

Equipped

```
1 Unequip  
2 OwnedCount -= 1  
3 Cash += SellValue  
4 RecalculateSpeed
```

Archive discovery remains.

52. Frozen Archive

Launch:

16 slots Prototype stretch:

8 slots Undiscovered:

??? Discovered: name; rarity; icon/visual. Selling all owned copies does not reverse discovery.

53. Follower Presentation

Follower is cosmetic. Recommended behavior:

- 1 Desired position behind player
- 2 -> smooth interpolation
- 3 -> snap closer when too far away

Do not use:

server-heavy pathfinding; Humanoid AI; follower position for gameplay calculations.

54. Pen Roaming Presentation

Optional cosmetic behavior:

- 1 Choose nearby valid point
- 2 -> move
- 3 -> pause
- 4 -> repeat

No gameplay effect. Can be removed without affecting economy.

55. Great Frost Player Rules

During Great Frost players may: remain outside; carry Ice; return home; train Warmth; reveal Ready Ice; manage creatures. Great Frost does not automatically: freeze; teleport; remove carried Ice.

56. Base Rules

Launch topology: up to 4 personal bases; grouped side-by-side along one shared home-camp edge; all face the same primary wilderness direction.

Each player has one assigned personal base. Warmth training occurs only in own Hearth/WarmZone. Current Warmth refill also occurs only in own WarmZone. Another player's WarmZone is visual/social space only for the visitor and provides no refill/training. Deposit occurs only at own Pen. Hearth upgrades only at own base. Pen management only at own base. Equip/Sell only at own base. Reveals occur at own Pen. Other players cannot manipulate another player's base progression.

Each base should contain one authored OnboardingIceSpawnPoint for the launch-only private onboarding exception.

57. Persistent Data - Launch

```
1    DataVersion = 1
2
3    Cash
4
5    MaxWarmth
6    HearthLevel
7
8    OwnedCreatures
9    PenSlots
10   EquippedCreature
11   DiscoveredCreatures
12
13   ThawingIce
```

Do not persist: CurrentWarmth; carried Ice; RevealSession; world position; Frozen state; follower position.

58. Join Defaults

New player:

```
1    Cash = 0
2
3    MaxWarmth = 70
4    CurrentWarmth = 70
5
6    HearthLevel = 1
7
8    OwnedCreatures = {}
9    PenSlots = {nil, nil, nil, nil}
10   EquippedCreature = nil
11
12   DiscoveredCreatures = {}
13   ThawingIce = {}
```

On every valid rejoin:

1 CurrentWarmth = MaxWarmth

Launch onboarding configuration: OnboardingIceEnabled = true OnboardingIceType = Regular
OnboardingIcePrivate = true OnboardingIceAffectedByGreatFrost = false

Eligibility is derived from existing state: DiscoveredCreatures is empty; ThawingIce is empty;
no carried Ice. No extra persistent onboarding boolean is required.

59. Data Operational Constants

Constant	Value
Autosave Interval	60 sec
Save Retry Attempts	3
Retry Backoff	1s / 2s / 4s
DataVersion	1

If loading fails:

never enter normal writable gameplay with a replacement blank profile that could overwrite legitimate data.

60. Warmth Network Update

Recommended:

Constant	Value
Warmth Simulation Step	0.25 sec or finer
Client Warmth Replication Target	4 updates/sec

Client may visually interpolate between updates. Gameplay truth remains server-side.

61. World Cycle Network Update

Server sends: phase; CycleIndex; transition timestamp.

Client displays countdown locally and periodically corrects. Do not send a server RemoteEvent every second merely to animate a countdown.

62. Launch Progression Timing Targets

These are playtest targets, not gates. First Creature

approximately 1-2 min First Hearth Upgrade

approximately 2-4 min Thick meaningful

approximately 5-10 cumulative min Ancient realistic

approximately 20-35 cumulative min Black

approximately 45-75 cumulative min Meteor

approximately 90-150 cumulative min Hearth Lv.10

longer-term progression target.

63. Prototype Timing Targets

First retrieval

under 60 sec First reveal

ideally 1-2 min First meaningful Equipped Speed

3-4 min First Great Frost

approximately 1:45 after the normal phase begins. First Hearth upgrade

approximately 2-4 min First Thick attempt

approximately 3-6 min Prototype pacing is intentionally compressed.

64. Low-Warmth Presentation

Safe

above 40%. Low

20-40%. Suggested: light frost; Warmth UI pulse. Critical

0-20%. Suggested: heavier frost; stronger wind;

warning audio; urgent UI. At zero:

server freezes player.

65. Ice Presentation Escalation

Regular Simple recognizable frozen block. Thick Larger / heavier-looking. Ancient Aged / prehistoric / fossil or runic visual language. Black Dark supernatural frost. Meteor Celestial / magical / spectacular. These are presentation profiles. They do not create additional gameplay systems.

66. Five Hero Moment Requirements

Hero 1 - First Great Frost Great Frost must be unmistakable even if effects are simplified.

Hero 2 - Impossible-Looking Ice At least one future opportunity should be visible substantially before comfortable access. No extra progression mechanic required.

Hero 3 - Barely-Made-It-Home Low/Critical Warmth presentation must escalate visibly.

Hearth must be readable from the return approach.

Hero 4 - Legendary Smash Legendary uses strongest reveal presentation. Do not increase required hits. Epicness comes from:

presentation + rarity + consequence.

Hero 5 - Endgame Speed Legendary first-pass:

WalkSpeed 32 Old regions should feel compressed by late-game traversal.

67. Special-Ice Hero Rule

Special Ice should create shared server attention. It must still require:

reach -> retrieve -> return -> thaw -> reveal. It never directly grants a creature.

68. Prototype Special Ice

Week 1:

```
1 PrototypeSpecialIceEnabled = false
```

P2 stretch only. Do not allow it to delay prototype P0/P1 validation.

69. Prototype Archive

Week 1:

OFF by default / P2 Internal discovery tracking may exist if trivial. Archive UI should not delay prototype completion.

70. Prototype Follower / Roaming

Minimal follower P1 if time permits. Follower polish P2. Pen roaming P2. None block prototype success.

71. Prototype Sell Scope

Minimum P1:

Sell from Inventory. Sell-from-Pen and Sell-Equipped may wait during Week 1. Launch must support all three.

72. Content Configuration Rule

Each creature should define at minimum:

```
1 CreatureID
2 DisplayName
3 Rarity
4 ModelName
5 SilhouetteProfile
```

Do not duplicate SellValue, PenIncome, or WalkSpeed per species unless an explicit later design change introduces species-specific overrides.

RarityConfig should define per rarity:

```
1 WalkSpeed
2 SellValue
3 PenIncomePerTick
4 RevealPresentationProfile
5 RarityUI
6 EffectsProfile
```

Economy and Speed derive from RarityConfig by default.

73. Ice Configuration Rule

Each Ice Type defines:

```
1 IceType
2 RarityWeights
3 CreaturePoolsByRarity
4 BaseThawDuration
5 CarryDrainMultiplier
6 PresentationProfile
```

Initial carry multiplier for all Ice:

1.75x Do not introduce different carry weights merely for flavor.

74. Hearth Configuration Rule

Each Hearth entry:

```
1 Level
2 UpgradeCost
3 WarmthTrainingRate
4 VisualStage
```

Do not create:

```
1 MaxWarmthCap
```

75. Region Configuration Rule

Each region:

```
1 RegionID
2 DisplayName
3 RecommendedWarmth
```

Do not create:

```
1 MinimumWarmthRequired
```

unless the design explicitly changes.

76. WorldCycle Configuration

Launch:

```
1 CycleLength = 300
2 GreatFrostDuration = 15
3 CycleEpochUnix = 0
4 ThawSurgePercent = 0.15
5 SpecialIceEveryNCycles = 3
6 GrabLockedDuringGreatFrost = true
```

Prototype:

```
1 CycleLength = 120
2 GreatFrostDuration = 15
3 GreatFrostThawMultiplier = 10
4 SpecialIceEnabled = false
5 GrabLockedDuringGreatFrost = true
```

Runtime-only:

```
1 LastProcessedFrostCycle
```

76A. SurvivalExperiment Configuration - Prototype Only

If the Day-5 experiment is enabled, keep its tunable values in one small prototype-only configuration surface such as:

```
1 SurvivalExperimentConfig
```

It may define only the values required by the approved experiment: - tool range/cooldown/damage; - Frost Sprite health/radius/telegraph/cooldown/projectile speed/damage; - temporary creature Vitality; - authored guard-profile counts; - conditional breakable hit counts.

Do not use this module as the seed of a generic combat/item/ability framework. If the experiment fails, it should be straightforward to disable/remove its config and implementation.

77. Structural State Invariants

Always preserve:

```
1 0 <= CurrentWarmth <= MaxWarmth
```

When the survival experiment is enabled:

```
1 0 <= CreatureVitality <= CreatureVitalityMax
2 CreatureVitality is session-only and applies only to the currently Equipped creature
3 Frosted/Downed never changes creature ownership
4 Frosted/Downed suppresses only the Equipped Speed bonus until Hearth recovery
```

```
1 MaxWarmth >= StartingMaxWarmth
```

```
1 0 <= carried Ice count <= 1
```

```
1 0 <= Equipped creature count <= 1
```

```
1 Active Pen creature count <= 4
```

1 PenCopies + EquippedCopies <= OwnedCopies

1 Each wilderness Ice has at most one OwnerUserID

1 Each ThawItem grants at most one creature

Carried raw Ice has no normal Inventory representation. There is no intentional DropIce action.

There is intentionally:

no gameplay maximum for logical ThawItems.

78. Configuration Values vs Structural Rules

Safe to tune Warmth drain; carry multiplier; refill; Hearth training rates; Hearth costs; Recommended Warmth; WalkSpeeds; creature values; rarity weights;

spawn counts; distances; thaw times; Great Frost duration; Frost bonus; Special cadence; Special rarity weights. Require explicit design decision Do not casually alter: one-Ice carry; uncapped MaxWarmth; Hearth = training rate; Equipped creature = Speed; 4 active Pen positions; unlimited logical thawing; 3-hit reveal; Great Frost world refresh; offline normal thawing; no hard area gates; one currency; server authority.

79. Initial Balance Philosophy

The intended progression feeling is: Early

“Getting that nearby Ice is an expedition.” Mid

“Regular is easy now. I want what is farther away.” Late

“I’m extremely fast, but the deepest Ice still gives me a reason to prepare.” Two independent progression axes:

MaxWarmth = how long I survive

Creature Speed = how much distance I cover during that time

80. Waiting Philosophy

Thaw timers should create:

anticipation not:

inactivity. While waiting, the player should usually be able to: retrieve another Ice; train Warmth; manage creatures; earn Cash; upgrade Hearth; wait for Great Frost. If players repeatedly have nothing useful to do:

first inspect the surrounding loop before automatically shortening every timer.

81. Great Frost Philosophy

The five-minute rhythm should encourage:

“I’ll stay for one more Frost.” Great Frost provides: 1. refreshed wilderness opportunities;

1. online thaw acceleration;
2. recurring spectacle;
3. periodic Special Ice. Do not convert it into a punishment timer.

82. Special Ice Philosophy

Special Ice should create:

urgency + aspiration + shared attention. It must remain rare enough that normal Ice stays meaningful. Initial cadence:

approximately every 15 minutes. Tune from real playtesting.

83. Creature Progression Philosophy

Higher rarity has three simple player-facing advantages: Pen

more passive Cash. Sell

more immediate Cash. Equip

more Movement Speed. Do not add additional creature stats without a separate design decision.

84. Production Scope Rule

If production falls behind, simplify: environmental art;

Hearth visual uniqueness; Pen roaming; follower polish; advanced silhouette presentation; VFX; Archive polish. Protect:

Retrieve -> Thaw -> Reveal -> Progress -> Retrieve again.

85. Final Canonical Configuration Summary

The first-pass launch build uses:

100 Starting MaxWarmth

4 Warmth/sec normal drain

7 Warmth/sec while carrying

uncapped MaxWarmth training

10 Hearth levels

16 / 20 / 24 / 28 / 32 movement progression

4 active Pen creatures

1-second aggregate Pen income interval

unlimited logical thawing Ice

1 / 3 / 7 / 15 / 25-minute thaw times

5-minute Great Frost cycle

15% online Thaw Surge

Special Ice every 3 Frost cycles

5 Ice types

16 creatures

4 rarities

1 currency

3-hit CRACK -> CRACK -> SMASH reveal These numbers are the values the launch implementation should start with. They are not permission for an AI assistant to reinterpret the underlying systems.

Current Week-1 prototype validation overrides additionally use: - 2.00x carry multiplier -> 8 Warmth/sec while carrying; - 16 / 22 / 28 / 34 None/Common/Uncommon/Rare WalkSpeed; - GreatFrostThawMultiplier = 10 only while the prototype Great Frost phase is active; - the Section 33A Survival Pressure constants only while the approved experiment is enabled.

Prototype experiment values are not launch defaults.

85A. v0.9 Revised Canonical Progression Summary

The current approved first-pass prototype direction additionally uses:

```
Per-copy CreatureGUID ownership
One inherited Ice WeightMultiplier per creature copy
0.75x-5.0x visual Scale
Scale = cbtr(WeightMultiplier)
IncomeMultiplier = Scale^2
SpeedMultiplier = 1 + 0.15 * (Scale - 1)
1-second aggregate Pen income
100 starting MaxWarmth
50% MaxWarmth/sec Hearth refill
Recommended Warmth: 100 / 1,000 / 10,000 / 100,000 / 1,000,000
Region Cold Multipliers: 1 / 10 / 100 / 1,000 / 10,000
Prototype Hearth training: 10 / 50 / 250 / 1,250 / 6,250 / 31,250 per second
```

These values are first-pass tuning values. The structural design is the inherited size jackpot + per-copy ownership + big-number Warmth progression. Balance may change without reverting that architecture.

86. Final AI Rule

When uncertainty exists:

Do not design by assumption. If the question is technical and does not change player behavior:

choose the smallest robust solution. If the question changes player behavior, progression, reward, risk, pacing, or the USP:

flag it before implementation. The AI's job while coding is:

to faithfully implement the current game design-not silently complete it with conventions from other games.